Matthew M. **Engelhard**

Assistant Professor of Biostatistics & Bioinformatics, Duke University · Machine Learning for Health

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Research Interests

Methods AI for clinical decision support; longitudinal risk surveillance; uncertainty quantification; causal methods

Applications electronic health records, digital health, mental health and neurodevelopment

Education

University of Virginia *Charlottesville, Virginia*

PhD in Systems and Information Engineering *Aug. 2011 - Dec. 2016*

- Dissertation: Adaptive Algorithms for Personalized Health Monitoring
- Advisor: Stephen D. Patek
- Committee: Peter A. Beling (chair), John C. Lach, Laura Barnes, Myla D. Goldman

University of Virginia *Charlottesville, Virginia*

MD *Aug. 2008 - Aug. 2014*

- Advanced Clinical Elective: Neurology
- Advisor: Myla D. Goldman

Duke University *Durham, North Carolina*

BSE in Biomedical and Electrical Engineering, BA in Mathematics *Aug. 2001 - May 2005*

Experience

Research Appointments

Department of Electrical & Computer Engineering, Duke University *Durham, North Carolina*

Assistant Professor *Nov. 2025 - present*

Duke AI Health, Duke University *Durham, North Carolina*

Director, Duke AI Health Data Science Fellowship Program *Aug. 2023 - May 2025*

Department of Biostatistics & Bioinformatics, Duke University School of Medicine *Durham, North Carolina*

Assistant Professor *Jan. 2021 - present*

- Division of Translational Biomedical Informatics
- Affiliations: AI Health, Center for Autism and Brain Development

Department of Psychiatry & Behavioral Sciences, Duke University School of Medicine *Durham, North Carolina*

Research Associate, Senior *Aug. 2017 - Dec. 2020*

- Conners/MHS Fellow in Digital Health, Duke ADHD Program
- School of Medicine Lead, +DataScience Program
- Affiliations: Center for Addiction Science and Technology, Lawrence Carin's Research Group, Center for Autism and Brain Development

Gates Foundation mHealth Engagement Lab, Praekelt.Org *Cape Town, South Africa*

Visiting Scholar *May 2017 - July 2017*

- Developed natural language processing and experiments for MomConnect, a digital maternal health platform

Department of Neurology, University of Virginia School of Medicine *Charlottesville, Virginia*

Research Associate *Jan. 2017 - May. 2017*

- Affiliations: Center for Wireless Health, Broadband Wireless Access and Applications Center, Multiple Sclerosis Clinic

Department of Systems and Information Engineering, University of Virginia *Charlottesville, Virginia*

Graduate Research Assistant *May 2011 - Dec. 2016*

- Affiliations: Center for Wireless Health, Broadband Wireless Access and Applications Center, Multiple Sclerosis Clinic

Department of Neurobiology, Duke University School of Medicine *Durham, North Carolina*

Undergraduate Research Assistant *May 2003 - May 2005*

- Affiliations: Laboratory of Miguel Nicolelis

Other Experience

National Security Agency *Fort Meade, Maryland*

Applied Research Mathematician *Jan. 2007 - Dec. 2007*

- Developed and implemented decryption algorithms for signals intelligence

Honeywell Technology Solutions, RF Section, US Naval Research Laboratory *Washington, District of Columbia*

Test Engineer *June 2006 - Dec. 2006*

- Developed procedures for testing radio frequency systems of tactical communications satellites

Publications

Refereed Articles

74. Cheng T, Gonzalez JM, **Engelhard MM**, Reed S, Ozdemir S. "Can LLMs Predict Patient Treatment Choices? A Discrete Choice Experiment Framework." Value in Health. 2026 May 5.
73. Hill ED, Loh DR, Davis NO, Goldstein BA, Dawson G, **Engelhard M**. "Early attention deficit hyperactivity disorder prediction from longitudinal electronic health records." Nature Mental Health. 2026 Apr 27:1-0.
72. Hong C, Niu M, Wang H, Wojdyla DM, Economou-Zavlanos N, **Engelhard MM**, Pignone M, Patel MR, Pencina MJ. "Performance of PREVENT Cardiovascular Risk in Electronic Health Record–Based Clinical Practice." JAMA Network Open. 2026 Apr 14;9(4):e266838.
71. Li F, Hill ED, Shu J, Gao J, **Engelhard MM**. "IRIS: Interpretable Retrieval-Augmented Classification for Long Interspersed Document Sequences." Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025.
70. Yan M, Xia M, Huang WA, Hong C, Goldstein BA, **Engelhard MM**. "Predicting Partially Observed Long-Term Outcomes with Adversarial Positive-Unlabeled Domain Adaptation." Proceedings of the 2025 Conference on Health, Inference, and Learning (CHIL), PMLR, 2025.
69. Zhao Y, **Engelhard MM**. "Balancing interpretability and flexibility in modeling diagnostic trajectories with an embedded neural Hawkes process model." Proceedings of the 2025 Conference on Machine Learning for Healthcare (MLHC), PMLR, 2025.
68. Sun M, **Engelhard MM**, Goldstein BA. "Borrowing From the Future: Enhancing Early Risk Assessment through Contrastive Learning." Proceedings of the 2025 Conference on Machine Learning for Healthcare (MLHC), PMLR, 2025.
67. Liu P, Shen Y, **Engelhard MM**, Goldstein BA, Pencina MJ, Economou-Zavlanos NJ, Zavlanos MM. "FairPOT: Balancing AUC Performance and Fairness with Proportional Optimal Transport." Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, 2025.
66. **Engelhard M**, Wojdyla D, Wang H, Pencina M, Henao R. "Exploring trade-offs in equitable stroke risk prediction with parity-constrained and race-free models." Artificial Intelligence in Medicine, 2025.
65. Wu P, Davis NO, **Engelhard MM**, Dawson G, Goldstein BA. "Using mixture cure models to address algorithmic bias in diagnostic timing: autism as a test case." JAMIA Open, 2025.
64. Wang M, **Engelhard MM**, Pun PH, Goldstein BA. "Discrete Time Neural Network Models to address time-varying predictor importance: An illustration in predicting mortality over different time horizons." IEEE Journal of Biomedical and Health Informatics, 2025.
63. Foote HP, Ou YJ, Bhatt S, **Engelhard MM**, et al. "Machine Learning Risk Prediction for Treated Retinopathy of Prematurity in Infants." Neonatology, 2025.
62. Wang D, Ownagh V, Hsu ST, Valikodath N, Kuo AN, **Engelhard M**, Vajzovic L. "Improving Pediatric Retinal Disease AI Screening Tools Constrained by Limited Data Using Synthetic Generation of Ultra-widefield Fundus Photos." Investigative Ophthalmology & Visual Science, 2025.
61. Sweitzer MM, Lazzari J, Lunsford-Avery J, McClernon FJ, Kollins SH, Perkins KA, **Engelhard M**. "Effects of initial nicotine exposure on cognition and nicotine reinforcement among non-smoking young adults with and without attention deficit hyperactivity disorder." Journal of Psychopharmacology, 2025.
60. Roberts AG, Patel R, Babu S, **Engelhard MM**, Greenberg RG, Ajmera A. "Can artificial intelligence pass the test? Evaluating chatbot scores on pediatric gastroenterology board-style questions." JPGN Reports, 2025.
59. Doerstling SS, **Engelhard MM**, Akrobetu D, et al. "Impact of Medical Conditions and Area Deprivation on Fundraising Success in Online Crowdfunding: Cross-Sectional Study." Journal of Medical Internet Research, 2025.
58. Loh DR, Hill ED, Liu N, Dawson G, **Engelhard MM**. "Limitations of Binary Classification for Long-Horizon Diagnosis Prediction and Advantages of a Discrete-Time Time-to-Event Approach: Empirical Analysis." JMIR AI. 2025 Mar 27;4:e62985.
57. Hill ED, Kashyap P, Raffanello E, Wang Y, Moffitt TE, Caspi A, **Engelhard M***, Posner J*. "Prediction of mental health risk in adolescents." Nature Medicine. 2025 Mar 5:1-7. (*co-senior authors)
56. Kumar A, Wang H, Muir KW, Mishra V, **Engelhard M**. "A Cross-Sectional Study of GPT-4–Based Plain Language Translation of Clinical Notes to Improve Patient Comprehension of Disease Course and Management." NEJM AI. 2025 Jan 23;2(2):AIoa2400402.
55. Wang D, Sklar B, Tian J, Gabriel R, **Engelhard M**, McNabb RP, Kuo AN. "Improving AI-based Microbial Keratitis Screening Tools Constrained by Limited Data Using Synthetic Generation of Slit Lamp Photos." Ophthalmology Science. 2024 Dec 20:100676.
54. Huang Y, McClernon FJ, and **Engelhard MM**. "Infinite hierarchical contrastive clustering for personal digital envirotyping." In: Findings of Machine Learning for Health (ML4H), December 2024.

53. Li B, Thomson AJ, **Engelhard MM***, Page D*. "On Neural Networks as Infinite Tree-Structured Probabilistic Graphical Models." Advances in Neural Information Processing Systems 37 (2024). (*co-senior authors)

52. Meng M, Li Y, Fuller M, Lanners Q, Habib A, Federspiel J, Quist-Nelson J, Shah S, Pencina M, Boggess K, Krishnamoorthy V, and **Engelhard M**. "Development and Validation of a Predictive Model for Maternal Cardiovascular Morbidity Events in Patients With Hypertensive Disorders of Pregnancy." Anesthesia & Analgesia. 2024 Nov 6:10.1213.

51. Huang WA, **Engelhard M**, Coffman M, Hill ED, Weng Q, Scheer A, Maslow G, Henao R, Dawson G, Goldstein BA. "A conditional multilabel model to improve prediction of a rare outcome: An illustration predicting autism diagnosis." Journal of Biomedical Informatics. 2024 Sep 1;157:104711.

50. Choi J, Palumbo N, Chalasani P, **Engelhard MM**, Jha S, Kumar A, Page D. "MALADE: Orchestration of LLM-powered Agents with Retrieval Augmented Generation for Pharmacovigilance." In: Machine Learning for Healthcare, August 2024

49. Sun M, **Engelhard MM**, Bedoya AD, Goldstein BA. "Incorporating informatively collected laboratory data from EHR in clinical prediction models." BMC Medical Informatics and Decision Making. 2024 Jul 24;24(1):206.

48. Lanners Q, Weng Q, Meng ML, **Engelhard MM**. "Common Event Tethering to Improve Prediction of Rare Clinical Events." In: The 40th Conference on Uncertainty in Artificial Intelligence, July 2024

47. Hickey J, Henao R, Wojdyla D, Pencina M, **Engelhard MM**. "Improving Event Time Prediction by Learning to Partition the Event Time Space." In: *International Conference on Artificial Intelligence and Statistics*. PMLR; 2024.

46. Song A, Lusk JB, Roh KM, Hsu ST, Valikodath NG, Lad EM, Muir KW, **Engelhard MM**, Limkakeng AT, Izatt JA, McNabb RP. "RobOCT-Net: Robotics and Deep Learning for Referable Posterior Segment Pathology Detection in an Emergency Department Population." *Translational Vision Science Technology*. 2024 Mar 1;13(3):12-.

45. Meng ML, Federspiel JJ, Fuller M, McNeil A, Habib AS, Quist-Nelson J, **Engelhard M**, Shah SH, Krishnamoorthy V. "Severe maternal morbidity according to mode of delivery among pregnant patients with cardiomyopathies." *Heart Failure.* 2023 Dec 1;11(12):1678-89.

44. Gu Z, Dawson G, **Engelhard M**. "Sex differences in the age of childhood autism diagnosis and the impact of co‐occurring conditions." *Autism Research*. 2023 Dec;16(12):2391-402.

43. Chen J, **Engelhard M**, Henao R, Berchuck S, Eichner B, Perrin EM, Sapiro G, Dawson G. "Enhancing early autism prediction based on electronic records using clinical narratives." *Journal of Biomedical Informatics*. 2023 Aug 1;144:104390.

42. Li B, Thomson AJ, **Engelhard MM**, Page D. "On Neural Networks as Infinite Tree-Structured Probabilistic Graphical Models." *arXiv preprint* arXiv:2305.17583. 2023 May 27.

41. Meng ML, Fuller M, Federspiel JJ, **Engelhard M**, McNeil A, Ernst L, Habib AS, Shah SH, Quist-Nelson J, Raghunathan K, Ohnuma T. "Maternal morbidity according to mode of delivery among pregnant patients with pulmonary hypertension." *Anesthesia Analgesia*. 2023 May 16:10-213.

40. **Engelhard MM**, Henao R, Berchuck SI, et al. "Predictive Value of Early Autism Detection Models Based on Electronic Health Record Data Collected Before Age 1 Year," *JAMA network open*, 2023;6(2):e2254303-e2254303.

39. **Engelhard MM**, Henao R. "Disentangling whether from when in a neural mixture cure model for failure time data," In *International Conference on Artificial Intelligence and Statistics, PMLR*, 2022:9571-9581.

38. Hong C, Pencina MJ, Wojdyla DM, **Engelhard MM**, et al. "Predictive Accuracy of Stroke Risk Prediction Models Across Black and White Race, Sex, and Age Groups," *JAMA*, 2023;329(4):306-317.

37. Doerstling SS, Akrobetu D, **Engelhard MM**, Chen F, Ubel PA. "A Disease Identification Algorithm for Medical Crowdfunding Campaigns: Validation Study," *Journal of Medical Internet Research*, 2022;24(6):e32867.

36. Bidopia T, **Engelhard MM**, Kollins SH, Lunsford-Avery JR. "Screen media technology and ADHD in children and adolescents: Potential perils and emerging opportunities," *Academic Press*, 2023

35. Glasgow TE, Adams EL, Ksinan A, **Engelhard MM**, et al. "Sleep onset, duration, or regularity: which matters most for child adiposity outcomes?," *International Journal of Obesity*, 2022;46(8):1502-1509.

34. **Engelhard MM**, D'Arcy J, Oliver JA, Kozink R, McClernon FJ, "Prediction of Smoking Risk From Repeated Sampling of Environmental Images: Model Validation," *Journal of medical Internet research*, 2021

33. Lunsford-Avery JR, Kollins SH, Kansagra S, Wang KW, **Engelhard MM**, "Impact of daily caffeine intake and timing on electroencephalogram-measured sleep in adolescents," *Journal of Clinical Sleep Medicine*, 2021

32. Lunsford-Avery JR, Wang KW, Kollins SH, Chung RJ, Keller C, **Engelhard MM**, "Regularity and Timing of Sleep Patterns and Behavioral Health Among Adolescents," *Journal of Developmental & Behavioral Pediatrics*, 2021

31. Subramanian V, **Engelhard M**, Berchuck S, Chen L, Henao R, Carin L, "SpanPredict: Extraction of Predictive Document Spans with Neural Attention," In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021

30. **Engelhard MM**, Navar AM, Pencina MJ, "Incremental Benefits of Machine Learning—When Do We Need a Better Mousetrap?," *JAMA cardiology*, 2021

29. Oliver JA, Sweitzer MM, **Engelhard MM**, Hallyburton MB, Ribisl KM, McClernon FJ, "Identifying neural signatures of tobacco retail outlet exposure: Preliminary validation of a "community neuroscience" paradigm," *Addiction biology*, 2021

28. Turpin M, Watson J, **Engelhard M**, Henao R, Thompson D, Carin L, Kirk A. "Machine Learning Prediction of Surgical Intervention for Small Bowel Obstruction," *medRxiv*, 2021

27. Vilardaga R, Fisher T, Palenski PE, Kumaresan V, Mannelli P, Sweitzer MM, McClernon FJ, **Engelhard MM**, Sabo PL, Garrison KA, "Review of popularity and quality standards of opioid-related smartphone apps," *Current addiction reports*, 2020

26. **Engelhard MM**, Berchuck SI, Garg J, Henao R, Olson A, Rusincovitch S, Dawson G, Kollins SH, "Health system utilization before age 1 among children later diagnosed with autism or ADHD," *Scientific reports*, 2020

25. Lunsford-Avery JR, Keller C, Kollins SH, Krystal AD, Jackson L, **Engelhard MM**, "Feasibility and acceptability of wearable sleep electroencephalogram device use in adolescents: Observational Study," *JMIR mHealth and uHealth*, 2020

24. **Engelhard MM**, Berchuck SI, D'Arcy J, and Henao R, "Neural Conditional Event Time Models," *Machine Learning for Healthcare (MLHC)*, 2020

23. **Engelhard MM**, Oliver JA, and McClernon FJ, "Digital envirotyping: quantifying environmental determinants of health and behavior," *npj Digital Medicine*, 2020

22. Lunsford-Avery JR, Damme KSF, **Engelhard MM**, Kollins, SH, and Mittal, VA, "Sleep/Wake Regularity Associated with Default Mode Network Structure among Healthy Adolescents and Young Adults," *Scientific Reports*, 2020

21. **Engelhard MM** and Kollins SH, "The Many Channels of Screen Media Technology in ADHD: a Paradigm for Quantifying Distinct Risks and Potential Benefits," *Current Psychiatry Reports*, 2019

20. **Engelhard MM**, Oliver JA, Henao R, Hallyburton M, Carin LE, Conklin C, and McClernon FJ, "Identifying Smoking Environments From Images of Daily Life With Deep Learning," *JAMA Network Open*, 2019

19. Raman S, **Engelhard MM**, and Kollins SH, "Driving the point home: novel approaches to mitigate crash risk for patients with ADHD," *Pediatrics*, 2019

18. Chen S, Perera R, **Engelhard MM**, Lunsford-Avery JR, Kollins SH, and Fuemmeler BF, "A Generic Algorithm for Sleep-Wake Cycle Detection using Unlabeled Actigraphy Data," *2019 IEEE EMBS International Conference on Biomedical & Health Informatics (BHI)*, 2019

17. Lunsford-Avery JR, **Engelhard MM**, Navar AM, and Kollins SH, "Validation of the Sleep Regularity Index in Older Adults and Associations with Cardiometabolic Risk," *Scientific Reports*, 2018

16. **Engelhard MM**, Xu H, Oliver JA, Hallyburton M, Carin L, and McClernon FJ, "Predicting Smoking Events with a Time-Varying, Semi-Parametric Hawkes Process Model," *Machine Learning for Healthcare (MLHC)*, 2018

15. **Engelhard MM**, Copley C, Watson J, Pillay Y, Barron P, and LeFevre A, "Optimizing mHealth Helpdesk Responsiveness in South Africa: Towards Automated Message Triage," *BMJ Global Health*, 2018

14. LeFevre AE, Dane P, Copley CJ, Pienaar C, Parsons AN, **Engelhard MM**, Woods D, Bekker M, Benjamin P, Pillay Y, and Barron P, "Unpacking the performance of a mobile health information messaging program for mothers (MomConnect) in South Africa: evidence on program reach and messaging exposure," *BMJ Global Health*, 2018

13. Dandu SR, **Engelhard MM**, Qureshi A, Gong J, Brant-Pearce M, Lach JC, and Goldman MD, "Understanding the Phsiological Significance of Four Inertial Gait Features in Multiple Sclerosis," *IEEE Journal of Biomedical and Health Informatics*, 2018

12. **Engelhard MM**, Patek SD, Lach JC, and Goldman MD, "Real-World Walking in Multiple Sclerosis: Separating Capacity from Behavior," *Gait & Posture*, 2018

11. Qureshi A, **Engelhard MM**, Brandt-Pearce M, and Goldman MD, "Demonstrating the Real-world Significance of the Mid-Swing to Heel Strike Part of the Gait Cycle Using Spectral Features," *14th International Conference on Wearable and Implantable Body Sensor Networks (BSN2017)*, Eindhoven, The Netherlands, May 2017

10. **Engelhard MM**, Patek SD, Lach JC, and Goldman MD, "Remotely Engaged: Lessons from Remote Monitoring in Multiple Sclerosis," *International Journal of Medical Informatics*, 2017

9. Qureshi A, Brandt-Pearce M, **Engelhard MM**, and Goldman MD, "Relationship between kernel density function estimates of gait time series and clinical data," *2017 IEEE International Conference on Biomedical and Health Informatics (BHI)*, Orlando, FL, Feb. 2017

8. **Engelhard MM**, Lach JC, Schmidt KM, Goldman MD, and Patek SD, "Adaptive Symptom Reporting for Mobile Patient-Reported Disability Assessment," *Proceedings of the 2016 Conference on Wireless Health*, Besthesda, MD, October 2016

7. **Engelhard MM**, Dandu SR, Patek SD, Lach JC, and Goldman MD, "Quantifying Six-Minute Walk Induced Gait Deterioration with Inertial Sensors in Multiple Sclerosis Subjects," *Gait & Posture*, 2016

6. Cincotta M, **Engelhard MM**, Stankey M, and Goldman MD, "Fatigue and Hydration Status in Multiple Sclerosis," *Multiple Sclerosis Journal*, 2016

5. Dandu SR, **Engelhard MM**, Goldman MD, and Lach JC, "Determining the Physiological Significance of Inertial Gait Features in Multiple Sclerosis," *2016 IEEE 13th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*, San Fracisco, CA, June 2016.

4. **Engelhard MM**, Schmidt KM, Engel CE, Patek SD, and Goldman MD, "The e-MSWS-12: Improving the Multiple Sclerosis Walking Scale using Item Response Theory," *Quality of Life Research*, 2016

3. **Engelhard MM**, Dandu SR, Lach JC, Goldman MD, and Patek SD, "Toward Detection and Monitoring of Gait Pathology using Rotation, Scale, and Offset Invariant Dynamic Time Warping," *BodyNets: 10th International Conference on Body Area Networks*, Sydney, Australia, September 2015

2. Gong J, **Engelhard MM**, Goldman MD, and Lach JC, "Correlations between Inertial Body Sensor Measures and and Clinical Measures in Multiple Sclerosis," *BodyNets: 10th International Conference on Body Area Networks*, Sydney, Australia, September 2015

1. Ribeiro S, Shi X, **Engelhard MM** et al., "Novel waking experience induces persistent sleep-dependent plasticity in the cortex but not in the hippocampus," *Frontiers in Neuroscience*, 2007

External Support

Current

Leveraging Artificial Intelligence to Predict Mental Health Risk among Youth Presenting to Rural Primary Care Clinics — *NIMH UF1*
Engelhard (co-PI, 30% effort), Posner (co-PI) — *2025 - 2029*

Digital Envirotyping to Develop Understanding of Cigarette smoking and the Environment — *NIDA R01*
McClernon (PI), Engelhard (co-I, 5% effort) — *2025 - 2030*

Machine Learning Methods and Deployment for Early Prediction of ASD and ADHD Risk from the Electronic Health Record — *NIMH K01*
Engelhard (PI, 50% effort) — *2022 - 2027*

Leveraging routinely collected health data to improve early identification of autism and co-occurring conditions — *NIMH P50*
Goldstein (PI), Engelhard (co-I, effort subsumed under K01) — *2022 - 2027*

The Gut Brain Parkinson's Disease Consortium (GBPDC): Coordinating and Data Management Center (CDMC) — *NIDDK U24*
Wruck (PI), Engelhard (KP, 2.5% effort) — *2024 - 2029*

Past

Evaluation of Cardiovascular Risk Prediction Equations across Diverse Sociodemographic Subgroups — *AHA*
Hong (PI), Engelhard (KP, 2% effort) — *2024 - 2025*

Improving stroke risk prediction with cohort data and machine learning methods — *NINDS R61/R33*
Pencina (PI), Engelhard (KP, 5-25% effort) — *2020 - 2025*

Sustainable Habits for Encouraging Even Teen Sleep (SHEETS): A Digital Intervention to Enhance Sleep Regularity and Psychiatric Health in Adolescents — *NIMH R34*
Lunsford-Avery (PI), Engelhard (co-I, 5% effort) — *2022 - 2025*

Cardiovascular Risk Prediction for Improved Maternal Health — *FAER*
Meng (PI), Engelhard (co-I, 0-10% effort) — *2022 - 2023*

Deep learning-based image analysis for assessing real-time smoking risk — *NIDA CEBRA R21*
McClernon (PI), Engelhard (co-I, 20% effort) — *2019 - 2021*

Novel Approaches to Infant Screening for ASD in Pediatric Primary Care — *NICHD R01*
Dawson (PI), Engelhard (KP, 10% effort) — *2019 - 2024*

Machine Learning Prediction of Adult ADHD Diagnosis — *Industry Sponsored*
Kollins (PI), Engelhard (co-I, 50% effort) — *2021*

Harnessing Sleep / Circadian Rhythm Data as a Biomarker to Mitigate Health Risks — *DIBS Research Incubator Award*
Lunsford-Avery (PI), Engelhard (co-I, 0% effort) — *2019 - 2020*

C. Keith Conners/MHS Fellowship in Digital Health — *Industry Sponsored Fellowship*
Engelhard (fellow, 100% effort) — *2017 - 2019*

Validation of a Motor Fatigue Measure in Multiple Sclerosis — *NINDS K23*
Goldman (PI), Engelhard (KP, 50% effort) — *2009 - 2013*

Broadband Wireless Access & Applications Center — *NSF I/UCRC*
Horowitz (PI), Engelhard (KP, 50% effort) — *2012 - 2015*

Teaching Experience

Courses Taught

2026-	**Deep Learning for Health Data (BIOSTAT 826)**, Dept. of Biostatistics & Bioinformatics, Duke University	*Durham, NC*
2019-2024	**Machine Learning for Health (CRP 278)**, Dept. of Biostatistics & Bioinformatics, Duke University	*Durham, NC*
2019-	**Applied Data Science (517CI)**, Dept. of Biostatistics & Bioinformatics, Duke University	*Durham, NC*

Courses Co-Taught

2019	**Machine Learning Methods & Practice (EGR190)**, Pratt School of Engineering, Duke Univ.	*Durham, NC*
2015	**Deterministic Decision Models**, Dept. of Systems & Information Engineering, Univ. of Virginia	*Charlottesville, VA*

Short Courses Co-Taught

2019	**+DataScience Lunch & Learn Series**, Duke University School of Medicine	*Durham, NC*
2019	**Duke/Duke-NUS Machine Learning School**, Duke University and Duke-NUS	*Singapore*
2019	**Machine Learning School for the School of Medicine**, Duke University	*Durham, NC*
2018-19	**Machine Learning Summer School**, Duke University	*Durham, NC*
2019	**Machine Learning Winter School**, Duke University	*Durham, NC*
2018-19	**+DataScience In-Person Learning Experiences**, Duke University	*Durham, NC*

Academic Presentations

Invited Lectures

32. "Medical data as the backbone of AI", 2026 AI in Clinical Medicine short course, Harvard Medical School, Virtual, June 2026.
31. "Toward interpretable, timely, context-sensitive clinical decision support", presented at the 2026 Duke Industry Statistics Symposium.
30. "Toward interpretable, timely, context-sensitive clinical decision support", presented as an Ashby Dialogue at the University of North Carolina, Greensboro, February 2026.
29. "Toward interpretable early autism surveillance", presented to the 2025 annual meeting of the NIH Autism Centers of Excellence, Virtual, June 2025.
28. "Toward interpretable, timely, context-sensitive clinical decision support", presented at the 2025 Duke-NUS AI + Medical Sciences Initiative (DAISI) Symposium, Singapore, April 2025.
27. "AI for improving patient comprehension of medical content: Results from a study of GPT4-based plain language translation of discharge summaries", presented at *NIH Collaboratory Grand Rounds*, March 2025.
26. "Toward interpretable early autism surveillance", presented to the Duke *Center for Autism and Brain Development* seminar series, October 2025.
25. "AI in Research", presentation at Duke LeadForward event for SoM leadership
24. "Building equitable AI-based clinical decision support systems", presented as a guest lecture for *Duke NURSING 726: Clinical Data Analytics*, March 2025.
23. "AI in Healthcare", presented at the *Duke National Health Research Conference*, February 2025.
22. "Synergistic use of predictive and generative AI for clinical decision support", presented to the *DCRI Pediatric GI Seminar Series*, January 2025.
21. "AI in medical education", presented to *Duke School of Medicine Education Grand Rounds*, December 2024.
20. "Research Applications with Harmonized Variables from the Framingham, MESA, ARIC, and REGARDS Studies", co-presented as a *Duke AI Health Virtual Showcase*, November 2024.
19. "ML Methods for Long-Horizon Event Prediction from EHRs", Presented as a lightning talk at the *Duke/Duke-NUS virtual symposium*, June 2024.
18. "ML Methods for Long-Horizon Diagnosis Prediction", Presented to the *Duke Center for Health Informatics*, April 2024.
17. "Methods for Long-Horizon Event Prediction from Electronic Health Record Data", presented to the *Duke Computational Biology and Bioinformatics Program Seminar Series*, March 2024
16. "What are machine learning and artificial intelligence, and why do they matter in health?", a *Duke AI Health Virtual Seminar*, October 2023
15. "Introduction to data science for global health", *Kilimanjaro Christian Medical Center*, Moshi, Tanzania, July 2023
14. "What are machine learning and artificial intelligence, and why do they matter in health?", a *Duke AI Health Virtual Seminar*, February 2023
13. "Predictive modeling with multi-modal health data", a *Duke AI Health Virtual Seminar*, November 2022

12. “Early prediction of neurodevelopmental risk from the electronic health record”, *Presented to the Duke Center for Applied Genomics and Precision Medicine Forum*, September 2021
11. “Introduction to Machine Learning for Clinical Medicine”, *Presented to the Duke School of Medicine Class of 2026*, Spring 2022
10. “Deep Learning for Medical Imaging Diagnostics”, *Presented to the Duke School of Medicine Class of 2026*, Spring 2022
9. “Introduction to Natural Language Processing for Clinical Text”, *Presented to the Duke School of Medicine Class of 2026*, Spring 2022
8. “Designing interpretable machine learning models for EHR-based Risk Prediction,” *Presented to the University of Virginia Center for Advanced Medical Analytics (CAMA)*, Charlottesville, VA, November 2021
7. “Early prediction of neurodevelopmental risk from the electronic health record,” *Center for Applied Genomics and Precision Medicine Forum*, Durham, NC, September 2021
6. “Knowing the Unknowable: Model Interpretability in EHR-based Risk Prediction,” *International Conference on Complex Acute Illness*, Virtual, August 2021
5. “Introduction to Machine Learning for Health,” *Guest Lecture, Duke Clinical Leadership Program (CLP212)*, Durham, NC, March 2020
4. “Deciphering Brain and Behavior with Data Science,” *Presented to the Duke University School of Medicine Class of 2023*, Durham, NC, January 2019
3. “Deep Learning for Medical Image Diagnostics,” *Presented to the Duke University School of Medicine Class of 2023*, Durham, NC, May 2019
2. “An Intuitive Approach to Healthcare Data Science,” *DukeAHEAD Health Professions Education Day*, Duke University School of Medicine, Durham, NC, September 2018
1. “Statistical Learning ‘On the Fly’: Applications in Maternal Health Messaging,” *Praekelt.Org lunch hour learning series*, Cape Town, South Africa, July 2017

Refereed Conference Presentations

10. **Engelhard M**, “Methods for Long-Horizon Event Prediction from Electronic Health Record Data”, Presented in a *2024 JSM Topic-Contributed Paper Session on Statistical Innovations in EHR Data Analysis*, August 2024.
9. Lunsford-Avery J and **Engelhard M**, “Light Regularity Associated with Cognitive Performance in Adolescents with ADHD”, Presented at SLEEP 2024, Houston, TX, June 2024
8. **Engelhard MM**, Vilardaga R, “Novel designs and analyses in smoking cessation mobile health research,” *2020 Annual Meeting of the Society for Research on Nicotine and Tobacco*, New Orleans, LA, March 2020
7. Jadhav S, **Engelhard MM**, Oliver JA, and McClernon FJ, “Improved prediction of environment-based smoking risk from images of daily life,” *2020 Annual Meeting of the Society for Research on Nicotine and Tobacco*, New Orleans, LA, March 2020
6. **Engelhard MM**, Berchuck SI, Garg J, Henao R, Rusincovitch S, Dawson G, and Kollins SH, “Patterns of Health Services Use before Age 1 in Children Later Diagnosed with ADHD,” *2020 Annual Meeting of the American Professional Society of ADHD and Related Disorders*, Washington, DC, January 2020
5. **Engelhard MM**, Lach JC, Schmidt KM, Goldman MD, and Patek SD, “Adaptive symptom reporting for mobile patient-reported disability assessment.” 2016 IEEE Conference on Wireless Health, National Institutes of Health, Bethesda, MD, August 2016
4. **Engelhard MM**, Dandu SR, Lach, JC, Goldman, MD, and Patek SD, “Toward detection and monitoring of gait pathology using inertial sensors under rotation, scale, and offset invariant dynamic time warping.” *10th EAI International Conference on Body Area Networks*, Sydney, Australia, September 2015
3. Gong, J, **Engelhard MM**, Goldman MD, and Lach JC “Correlations between inertial body sensor measures and clinical measures in multiple sclerosis,” *10th EAI International Conference on Body Area Networks*, Sydney, Australia, September 2015
2. **Engelhard MM**, Goldman MD, Lach JC, and Patek SD, “Characterizing MS Disability with Accelerometer-Based Motion Capture in the Field,” *2013 National Library of Medicine Informatics Training Conference*, Salt Lake City, UT, 2013
1. **Engelhard MM** and Learmonth G, “A Participatory Simulation for Health Care Policy Analysis,” *2012 National Library of Medicine Informatics Training Conference*, Madison, WI, 2012

Refereed Conference Posters

21. Pisetsky D, Zhao Y, **Engelhard M**, Eudy A, Tedeschi P, Verdone A, Clowse M, Criscione-Schreiber L, Doss J, Maheswaranathan M, Sadun R. “A Novel Modeling Approach to Elucidate the Role of Autoantibodies in Complement Activation in SLE.” In ARTHRITIS RHEUMATOLOGY 2024 Sep 1 (Vol. 76, pp. 4860-4863).
20. Lunsford-Avery J, Krystal A, **Engelhard M**. “Light regularity associated with cognitive performance in adolescents with ADHD.” In SLEEP 2024 May 1 (Vol. 47, pp. A349-A349).
19. Loh DR, Hill E, Navar AM, Dawson G, and **Engelhard MM**. “Development of EHR-based Neurodevelopmental Surveillance Dashboard for Clinicians.” In: Machine Learning for Healthcare (MLHC), August 2024
18. Weng Q, Lanners Q, **Engelhard M**, “Conditions for Enhanced Rare Event Prediction with Multi-Label Learning.”, *2024 AMIA Informatics Summit*, Boston MA, March 2024

17. Tan X, Huang Y, Oliver J, McClernon FJ, **Engelhard M**, "Linking Home Environments and Psychosocial Health through Computer Vision.", *2024 AMIA Informatics Summit*, Boston MA, March 2024

16. Tan X, Rubenstein D, Oliver J, McClernon FJ, **Engelhard M**, "Characterizing Home Environments of Smokers and Nonsmokers using Computer Vision.", *2024 Annual Meeting of the Society for Research on Nicotine and Tobacco*, Edinburgh, Scotland, March 2024

15. Falls A, **Engelhard M**, Lunsford-Avery J, "Light regularity associated with sleep regularity and timing in adolescents with and without ADHD.", *2021 Annual Meeting of the American Professional Society of ADHD and Related Disorders*, Orlando, FL, January 2024

14. Lunsford-Avery J, Kansagra S, Kollins S, **Engelhard M**. "Leveraging convenient wearable technology to assess adolescent sleep: Physical and psychiatric health correlates of single-channel sleep EEG in a community sample of youth," *Poster presented at SLEEP 2022*.

13. Oliver J, **Engelhard M**, Stevens B, McQuoid J, McClernon FJ. "Probing Disparities in Exposure to Smoking Contexts Using Computer Vision," *Poster presented at the American College of Neuropsychopharmacology Annual Meeting*, 2022:439-440.

12. **Engelhard M**, McGough D, Kollins S, "Improving the Predictive Value of Self-Report in Adult ADHD Diagnosis," *68th Annual Meeting of the American Academy of Child & Adolescent Psychiatry*, Virtual, October 2021

11. **Engelhard M**, Berchuck S, Garg J, Rusincovitch S, Dawson G, Kollins S, "Patterns of Health Services Use Before Age 1 in Children Later Diagnosed With ADHD," *2021 Annual Meeting of the American Professional Society of ADHD and Related Disorders*, Virtual, January 2021

10. **Engelhard MM** and Kollins SH, "Predicting ADHD-Related Symptoms from Free-Text Responses to Internally and Externally Focused Questions," *2020 Annual Meeting of the American Professional Society of ADHD and Related Disorders*, Washington, DC, January 2020

9. **Engelhard MM**, Oliver J, Henao R, Hallyburton M, Carin L, Conklin C, and McClernon FJ, "Identifying Smoking-Risk Environments from Everyday Images with Deep Learning", *2019 Annual Meeting of the Society for Research on Nicotine and Tobacco*, February 2019

8. Oliver JA, **Engelhard MM**, Hallyburton M, Ribisl K, and McClernon FJ, "Differential Exposure to Tobacco Retail Outlets among Smokers and Non-Smokers", *2018 Annual Meeting of the Society for Research on Nicotine and Tobacco*, Feburary 2018

7. Brenton JN, **Engelhard MM**, Woolbright E, Koshiya H, Herod SG, Engel CE, and Goldman MD, "Continuous accelerometry as a measure of physical activity impairment in paediatric-onset multiple sclerosis subjects versus healthy controls," *Multiple Sclerosis Journal*, 2017

6. Koshiya H, Herod SG, Engel CE, **Engelhard MM**, Goldman MD, Brenton JN, "Utility of Physical Disability Outcome Measures in Pediatric-Onset Multiple Sclerosis," *American Academy of Neurology 69th Annual Meeting*, Boston, MA, April 2017

5. **Engelhard MM**, Schmidt KM, Patek SD, Lach JC, and Goldman MD, "An Item Response Theory Analysis of the Multiple Sclerosis Walking Scale," *2015 Consortium of Multiple Sclerosis Centers Annual Meeting*, Dallas, TX, May 2015

4. Zee RS, Seaman SJ, **Engelhard MM**, Gong J, and Schenkman NS, "Surgical Attention and Movement in Novice and Expert Robotic Surgeons," *American Urological Association Annual Meeting*, Orlando, FL, USA, May 2014

3. **Engelhard MM**, Patek SD, Lach JC, and Goldman MD, "Detecting changes in gait characteristics during a six-minute walk using wireless technology in MS subjects," *3rd International Symposium on Gait and Balance in Multiple Sclerosis*, St. Louis, MO, 2013

2. **Engelhard MM**, Ribeiro S, Shi X, et al., "Sensory experience induces long-lasting changes in cortical neuronal firing rates during slow-wave sleep," *2004 Society for Neuroscience Annual Meeting*, San Diego, CA, 2004

1. Shi X, **Engelhard MM**, Ribeiro S, et al., "Experience-Dependent zif-268 Expression in the Sensory Cortex across Slow-Wave, Intermediate, and REM Sleep," *2004 Society for Neuroscience Annual Meeting*, San Diego, CA, 2004

Other Presentations and Posters

5. **Engelhard MM**, Patek SD, Lach JC, and Goldman MD, "Wireless, Remote Monitoring of Patients with Multiple Sclerosis: New Results and Upcoming Studies," *2015 Broadband Wireless Access and Applications Center Spring Meeting*, South Bend, IN, 2015

4. **Engelhard MM**, Sheridan K, Patek SD, Lach JC, and Goldman MD, "Toward Remote, Adaptive Disability Tracking using Wireless Technology in Multiple Sclerosis Subjects," *2014 Broadband Wireless Access and Applications Center Fall Meeting*, Charlottesville, VA, 2014

3. **Engelhard MM**, Sheridan K, Patek SD, Lach JC, and Goldman MD, "Accuracy and Clinical Utility of Wireless Personal Health Monitoring Devices in Multiple Sclerosis," *2014 Broadband Wireless Access and Applications Center Spring Meeting*, Auburn, AL, 2014

2. **Engelhard MM**, Sheridan K, Patek SD, Lach JC, and Goldman MD, "Detecting Changes in Gait Characteristics in MS Subjects using Wireless Inertial Sensors," *2013 Broadband Wireless Access and Applications Center Fall Meeting*, Tucson, AZ, 2013

1. **Engelhard MM**, Sheridan K, Patek SD, Lach JC, and Goldman MD, "New Tools for Clinician and Patient Centered Management of Health Data," *2012 Wireless Access and Applications Center Fall Meeting*, Charlottesville, VA, 2012

Honors & Awards

2022	**Early Career Reviewer (ECR),** Center for Scientific Review, National Institutes of Health	*Bethesda, MD*
2017	**Outstanding Graduate Student Award,** Dept. of Systems and Information Engineering, Univ. of Virginia	*Charlottesville, VA*
2015	**Member,** Omega Rho International Honor Society for Operations Research	*Charlottesville, VA*
2015	**Teaching Fellow,** School of Engineering and Applied Sciences, University of Virginia	*Charlottesville, VA*
2014-16	**Skinner Scholar,** St. Paul's Memorial Church	*Charlottesville, VA*
2014	**Research Scholar,** Foundation for the Consortium of Multiple Sclerosis Centers	*Hackensack, NJ*
2011-13	**T15 Training Program Fellow,** National Library of Medicine	*Bethesda, MD*
2005	**Member,** Tau Beta Pi Engineering Honor Society	*Durham, NC*
2001	**FOCUS Program,** Exploring the Mind, Duke University	*Durham, NC*

Professional Activities

Grant Review

NIH review panels: Child Psychopathology and Developmental Disabilities (CPDD), February 2025; Clinical Informatics and Digital Health (CIDH), September 2023; Topics in Clinical Informatics and Data Analytics (special emphasis panel), July 2023; Clinical Data Management and Analysis (CDMA), October 2022

Ad Hoc Scientific Review

Association for Computational Linguistics (ACL) Rolling Review (ARR); Conference on Uncertainty in Artificial Intelligence (UAI); International Conference on Machine Learning (ICML); Conference on Neural Information Processing Systems (NeurIPS); Conference on Health, Inference, and Learning (CHIL); International Conference on Artificial Intelligence and Statistics (AISTATS); AAAI Conference on Artificial Intelligence; PLOS One; npj Digital Medicine; Scientific Reports; Journal of Child Psychology and Psychiatry; IEEE Transactions on Biomedical Engineering; IEEE Journal of Biomedical and Health Informatics; BMJ Global Health; Multiple Sclerosis Journal; Sensors

Organization Membership

American Medical Informatics Association, American Professional Society for ADHD and Related Disorders, Society for Research on Nicotine and Tobacco

Committee Membership

2024-25	**Computational Biology and Bioinformatics PhD Admissions Committee**, Department of Biostatics and Bioinformatics, Duke University School of Medicine	*Durham, NC*
2023-26	**Biostatistics PhD Admissions Committee**, Department of Biostatics and Bioinformatics, Duke University School of Medicine	*Durham, NC*
2021, 2023	**TBI Faculty Search Committee**, Department of Biostatics and Bioinformatics, Duke University School of Medicine	*Durham, NC*
2019-20	**Data Infrastructure Working Group**, Dept. of Psychiatry & Behavioral Sciences, Duke University SoM	*Durham, NC*
2018	**Curriculum Reform Initiative**, Committee for Innovation in Clinical Education, Duke University SoM	*Durham, NC*

Student Advisees

PhD	Liwen Sun (Duke Biostat), Pengxi Liu (Duke ECE), Fengnan Li (Duke CBB), Qin Weng (Duke CBB), Yuankang Zhao (Duke Biostat), Yuqi Li (Duke Biostat), Scott Sun (Duke Biostat, co-Advisor)
Master's	Dhruv Menon, Haoyu Gong (Duke BME MEng); Bingkun Wang (Duke ECE MEng); Isabella Wang (Duke CS MEng); Ya-Yun Haung, Jiaxin (Cindy) Gao, Samantha (Xiangyu) Wang (Duke MIDS); Wei Ding, Mai Li, Arda Mert, Julia Shoobe, Max Freitas, Zhiyuan (Allen) Gao, Fengnan Li, Haoyuan (Luke) Wang, Jason Ma, Qin Weng, Xin Tan, Xi Liang, Bryce Phares, Zhengyi (Gloria) Gu, Yuqi (Lucilla) Li, Yiyuan Yao (Duke MB)
Medical	Roshni Varma (Duke SoM '27), Anivarya Kumar (Duke SoM '25), Loh De Rong (Duke-NUS '25)
Undergraduate	Matthew Xie (Duke '29), Heman Burre (Duke '26), Shayan Kothari (Duke '25), Abhishek Jadhav (Duke '21), Miles Turpin (Duke '20)
Other	Alice Thiers (Summer 2026 Visiting Intern from École Normale Supérieure (ENS) Paris-Saclay)

Skills

Machine Learning	Python, Pytorch, Tensorflow, Pandas, Jupyter, SQL
Statistical Software	Matlab, R
Other Programming	C, Perl, HTML/CSS, PHP, Javascript
Other Software	Git, LaTeX, Markdown